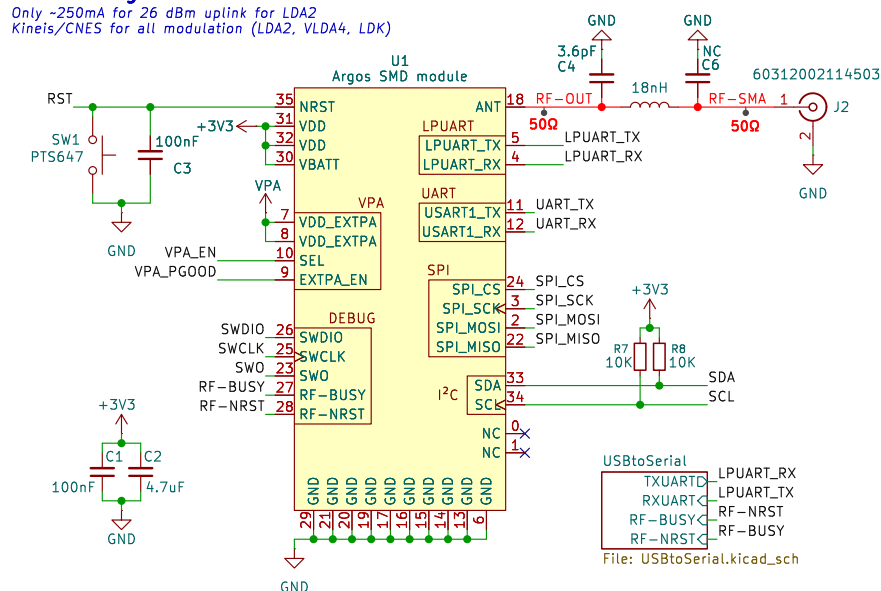


Arribada Argos SMD module

Only ~250mA for 26 dBm uplink for LDA2
Kineis/CNES for all modulation (LDA2, VLDA4, LDK)



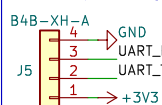
Connecteur N°1

I²C Optional signals



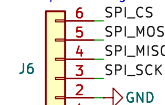
Connecteur N°2

LPUART Optional signals



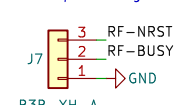
Connecteur N°3

SPI Optional signals B6B-XH-A



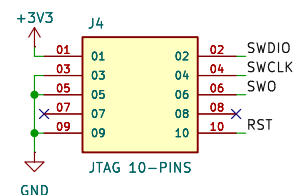
Connecteur N°4

ARGOS Optional signals

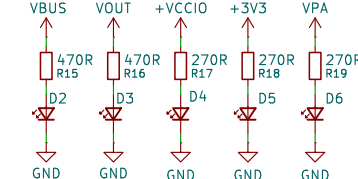


Serial Wire Debug

ARM Cortex 10pin Debug Connector

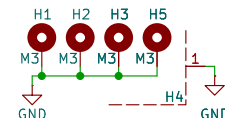


Power supply status



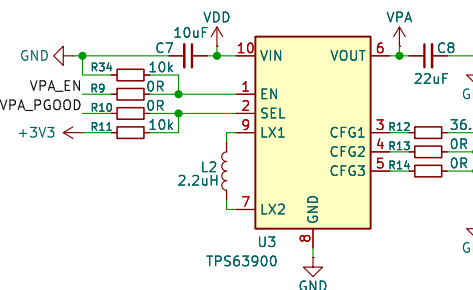
Mechanical and shielding

Shield 36102131265
M3 holes referenced to ground



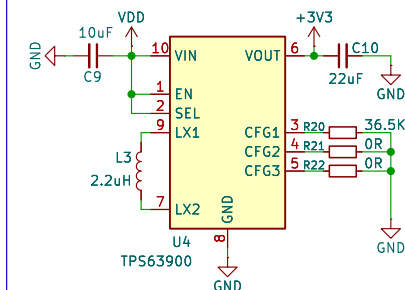
External Power Amplifier

Buck-boost Converter



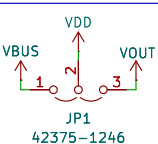
Supply power 3V3

Buck-boost Converter



Power select

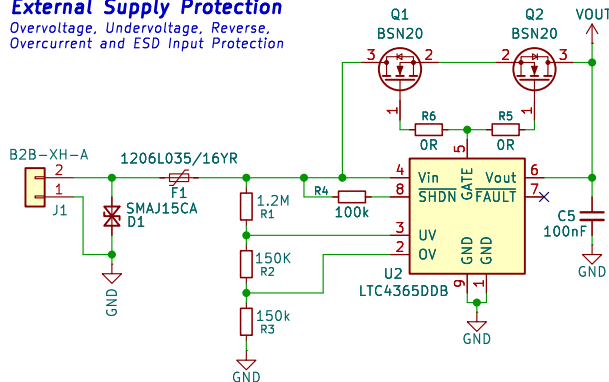
Jumper 2.54mm
Common 3V3 supply
Power amplifier 3V3 supply



Select the 5V power supply from the USB or external connector for the PA power supply and 3V3 for the Argos SMD.

External Supply Protection

Overvoltage, Undervoltage, Reverse, Overcurrent and ESD Input Protection



The LTC4365 is configured using an external resistor divider sized according to the manufacturer's recommended topology, providing defined undervoltage and overvoltage thresholds at 2.5 V and 5.0 V respectively. The divider values are calculated assuming an allowable UV pin offset of VOS(UV) = 3 mV, in order to minimize threshold errors due to leakage currents. Additional input protection is implemented using a resettable PTC fuse (1206L035YR) to limit fault current, and a bidirectional TVS diode (SMAJ15CA) to protect against ESD and transient overvoltage events on the input supply.

Centre National d'Etudes Spatiales

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Tél: +33561273131
Auteur: MIALHE PAUL

Sheet: /
File: ARGOS DEMO.kicad_sch

Title: ARGOS DEMO

Size: A4 Date: 2026-01-21

KiCad E.D.A. 9.0.2-238-gb1649071e8

Rev: V°1.2

Id: 1/2

FT230XS USB↔UART Bridge

USB 2.0 Full-Speed (12 Mbps)
UART Data transfer rates from 300 baud to 3 Mbaud at TTL levels

Function Summary :

- Converts USB 2.0 Full-Speed signaling to asynchronous UART
- Enumerates as a virtual COM port on the host (FTDI VCP)
- Provides TXD / RXD UART interface at 3.3 V logic
- USB engine + clock fully integrated (no crystal required)
- 3.3 V regulator onboard (3V3OUT), VCCIO externally selectable (3.3 V in your design)
- Supports CBUS programmable pins:
 - CBUS1 = TXLED#
 - CBUS2 = RXLED#
 - CBUS0/3 = connector

VBUS Protection & Filtering :

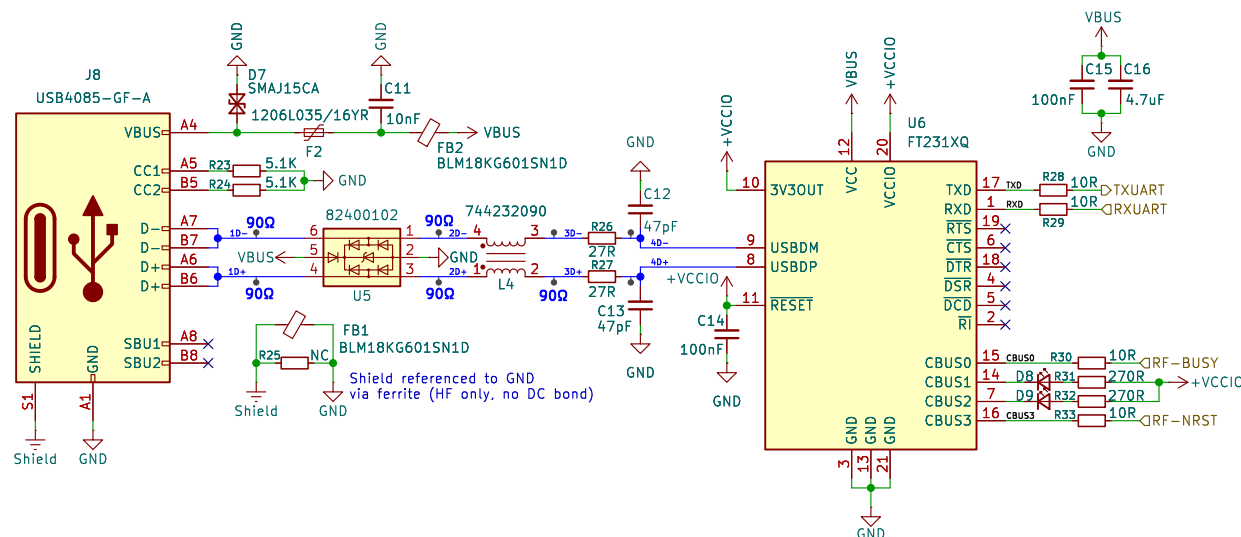
- PTC Fuse: 1206L035YR / 1206L050YR (I_{hold} 0.35A / 0.50A, 1206 footprint)
- TVS VBUS: SMAJ15CA, bidirectional (SMA / DO-214AC package)
- VBUS EMI Filter: BLM18KG601SN1D (600Ω @100MHz, 0603)
- Decoupling: 10μF + 100nF (placed immediately after ferrite)

USB Data Lines (D+ / D-)

- Common-Mode Choke: 744232090 (Würth Elektronik)
- High-Speed ESD Array: WE-TVS-82400102
- Series Termination: 27Ω on D+ and D- (matched pair)

USB Shield

- Shield → GND via BLM18KG601SN1D (same ferrite as VBUS, 600Ω @100MHz)



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Sheet: /USBtoSerial/

File: USBtoSerial.kicad_sch

Title: USB ↔ UART Bridge

Size: A4

Date: 2025-12-07

KiCad E.D.A. 9.0.2-238-gb1649071e8

Rev: V°1.2

Id: 2/2